Quantum-inspired dynamical models on quantum and classical annealers

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Introduction

We propose a practical, physics-inspired benchmarking suite to challenge both quantum and classical computers by mapping real-time quantum dynamics to a common optimization format.

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Quantum Annealing

Initialize in the ground state of simple driver Hamiltonian:

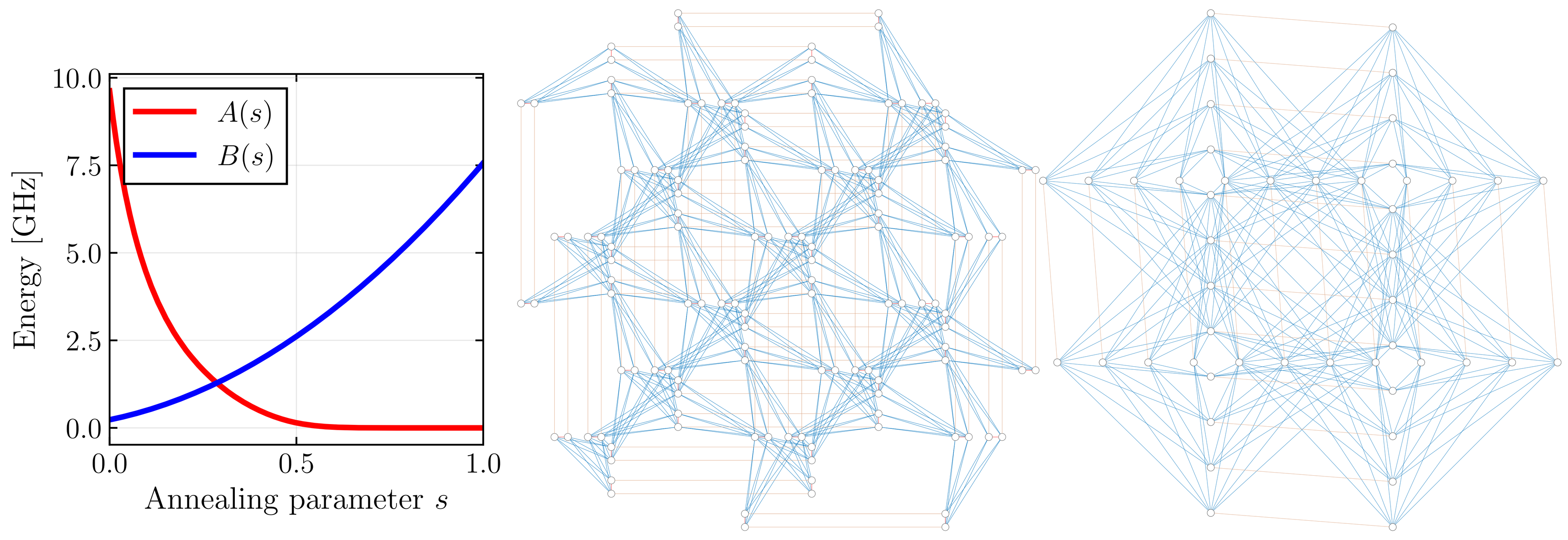
$$|+\rangle^{\otimes N} = \frac{1}{\sqrt{2^N}} \sum_{x \in \{0,1\}^N} |x\rangle$$

with N being the number of qubits. The goal is to find the ground state of a problem Hamiltonian

$$H_P = \sum_{i=1}^{N} h_i \widehat{\sigma}_z^{(i)} + \sum_{i<j} J_{ij} \widehat{\sigma}_z^{(i)} \widehat{\sigma}_z^{(j)},$$

During the annealing process, the system evolves under a time-dependent Hamiltonian

$$H_{\text{Ising}} = \frac{A(s)}{2} H_D + \frac{B(s)}{2} H_P,$$



(a) (b) (c)

Figure: (a) Time-dependent energy scales executed by the D-Wave quantum annealer. (b) Native connectivity map for the Pegasus architecture (D-Wave Advantage). (c) Connectivity map for the newer Zephyr architecture (D-Wave Advantage2).

Methodology

1. Data Collection.
2. Signal Processing.
3. Result Verification.

Conclusion

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